

WT Experiment 3

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AIM : Wireless Notice Board using Bluetooth

Introduction

The Wireless Notice Board using Bluetooth and an Arduino Uno is an innovative IoT-based project that allows users to send messages to a notice board wirelessly using a mobile phone via Bluetooth. This project involves integrating an LCD display with a temperature sensor, where the default display shows the temperature data, and users can send their messages through Bluetooth to be displayed on the board. This system leverages Bluetooth communication to wirelessly transfer data, making it a simple and effective solution for creating smart notice boards.

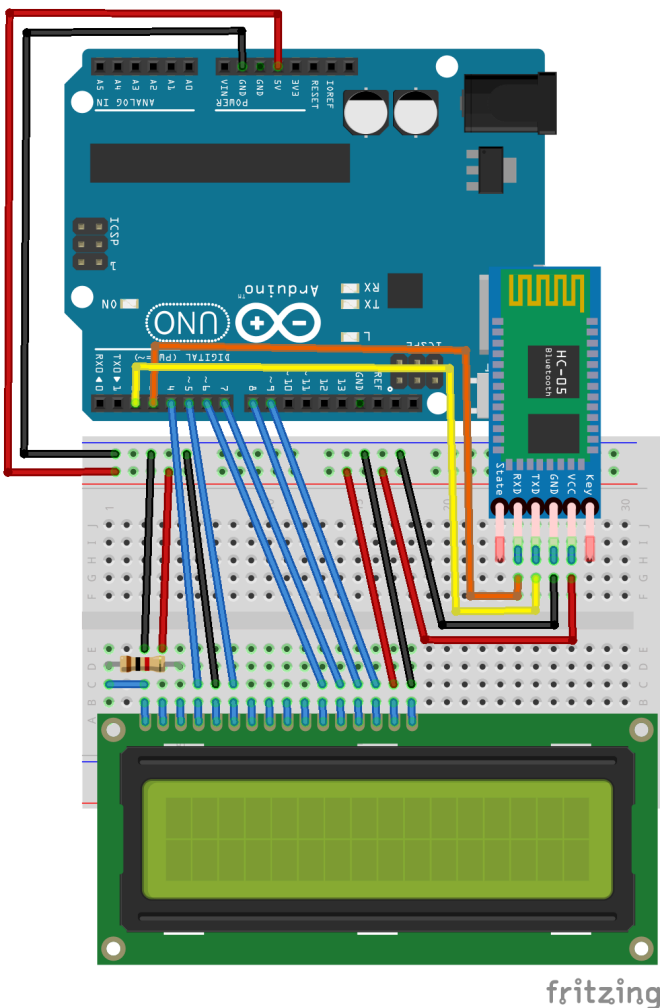
Key Features

- **Wireless Communication:** The project utilizes Bluetooth communication to send messages from a mobile device to the Arduino.
- **LCD Display:** The 16x2 LCD display shows the default temperature reading, and users can send text messages to be displayed.
- **Temperature Monitoring:** The system continuously monitors and displays the temperature in real-time using a temperature sensor.
- **Mobile Control:** Bluetooth connectivity allows users to control the display from a mobile phone via a Bluetooth app.
- **Simple and Interactive:** The system is easy to interact with, requiring no physical intervention to display messages.
- **Low Power Consumption:** The use of Arduino and Bluetooth technology ensures low power consumption, making it ideal for long-term use.

Working Principle

1. **Bluetooth Communication:** The HC-05 Bluetooth module is used for wireless communication between the mobile device and the Arduino. The mobile device sends data to the Bluetooth module, which is received by the Arduino.
2. **Arduino Uno:** The Arduino Uno microcontroller receives the data from the Bluetooth module and controls the LCD display accordingly. It also processes the input from the temperature sensor.
3. **LCD Display:** The 16x2 LCD display shows the default temperature value from the temperature sensor. When a message is sent via Bluetooth, it overrides the temperature display and shows the received message.
4. **Temperature Sensor:** The temperature sensor (such as a DHT11 or LM35) measures the ambient temperature and provides real-time data to the Arduino for display.

Circuit Diagram:



Advantages

- **Wireless Control:** No physical connection is needed between the notice board and the user, offering convenience and flexibility.
- **Real-Time Data:** The system provides real-time temperature data, which can be valuable in many applications, such as classrooms or offices.
- **Energy Efficient:** The system uses low-power components, making it energy-efficient and cost-effective for continuous operation.
- **Easy to Build:** The components are easily available and the setup is relatively simple, making it ideal for a beginner-level IoT project.
- **Interactive:** The system provides an interactive experience where users can send messages or notices from their mobile phones.

Components Required

- **Arduino Uno:** The main microcontroller for controlling the system.
- **HC-05 Bluetooth Module:** For wireless communication between the mobile device and the Arduino.
- **16x2 LCD Display:** To display messages and temperature readings.
- **Temperature Sensor (DHT11 or LM35):** To measure the temperature.
- **1-ohm Resistor:** For current limiting in some components (such as the Bluetooth module or sensor).
- **Jumper Wires:** For connecting components on the breadboard.
- **Half Breadboard:** To connect and arrange the components.
- **Mobile Phone with Bluetooth:** To send messages via Bluetooth.

Conclusion

The Wireless Notice Board project is an excellent demonstration of how IoT technology can be integrated into everyday life. Using Bluetooth communication and Arduino, it offers a simple yet effective solution for displaying both real-time sensor data (such as temperature) and custom messages. This project provides an introduction to wireless communication, microcontroller programming, and sensor interfacing, which are valuable skills in the field of IoT. With its low power consumption and ease of use, it is an ideal project for both learning and practical applications, such as smart classrooms, offices, or public spaces.